

font test

Polynomials

1. Hence write $p(x)$ as a product of linear factors. [3 marks]

2. The polynomial p is such that $p(x) = 6x^3 + x^2 - 12x + 5$. Find the remainder when $p(x)$ is divided by $x - 2$. [1 mark]

3. Using answers to part (a), find remainder when $p''(x)$ is divided by x [2 marks]

4. On the axes, sketch the graph of $y = 3(x - 3)(x - 1)(x + 2)$, stating the intercepts with the coordinate axes. [3 marks]

5. The polynomial $p(x) = mx^3 - 29x^2 + 39x + n$, where m and n are constants, has a factor $3x - 1$, and remainder 6 when divided by $x - 1$. Show that $x - 2$ is a factor of $p(x)$. [6 marks]

6. Find $p(x)$ in the form $(2x-5)q(x)+r$, where $p(x)=6x^3-35x^2+34x+45$, $q(x)$ is a polynomial and r is a constant. [3 marks]

7. Hence write the expression $p(x)-5$ as a product of linear factors. [2 marks]

Answer key

1. $p(x) = (2x - 1)(3x + 5)(x - 1)$
2. 33
3. Remainder = 4
4. Cubic with roots at -2, 1, 3 and y-intercept at 18
5. $m = 6$, $n = -10$; $p(2) = 0$
6. $q(x) = 3x^2 - 10x - 8$, $r = 5$
7. $p(x) - 5 = (3x + 2)(x - 4)(2x - 5)$